

EMBEDDED & EDGE ARCHITECTURES

Embedded & Edge Architectures track is a large-scale diploma program designed to take engineers from embedded systems concepts through to advanced, industry-grade specializations in automotive, real-time operating systems, Embedded Linux, Embedded Android, edge/Physical AI, and Embedded IOT. The track combines structured lectures with extensive hands-on XP.

JOB PROFILES

Embedded Kernel S/W Engineer · Embedded S/W Engineer · QNX ES - Engineer · IVI S/W Engineer · AUTOSAR S/W Engineer · Emb-Linux Engineer/yocto developer · ES UI/HMI Dev. (C++/Qt) · Edge/physical AI ES Engineer · ROS Dev. for Autonomous vehicle



TOTAL HOURS

1,365hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



Where graduates work

Brightskies

VehicleEvo

SEITech

CyShield

ALARABY

Valeo

Coretech

Luxoft

TechConsultix

Si-Vision

Appraid

expleo

eJad



Selection process

01 IQ & English Exam



Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam



Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview



Interview with the technical committee.

04 Interpersonal Interview



Candidates who pass the technical interview are invited to this final stage.



Who may apply

Open to graduates in Electrical power Engineering, Electro-mechanical Engineering, Electronics Engineering, Communication Engineering, Mechatronics Engineering, Computer Engineering, computer science with a minimum grade of Good or higher.

Program content – 1,365 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Core Courses	379 hrs	28%	
● Advanced Core Courses	587 hrs	43%	Edge/Physical AI, QNX, IVI, Multi-Core
● Soft Skills	99 hrs	7%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor, Progressive Teamwork (Workshop), Best Practices For Remote Working (Workshop)
● Graduation Project	300 hrs	22%	

Hands-on labs/Projects are conducted in-person.

Student Deliverable

Every graduate must complete a hands-on project that demonstrates their practical skills, technical expertise, and comprehensive understanding of the core principles of embedded systems.

Before you apply

Explore these Topics before your interview.

- C concepts
- AVR interfacing
- Explored datasheets of any microcontroller
- Basics of electronics
- Computer architecture
- OOP concepts

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.